



Manufacturers of Food Machinery

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CHINS 100

EQUIPMENT MANUAL

for

UTZ Foods

Masa Pre-Sheeter

Job # 090101 SO# 050111

CONTENTS

NOTE: The following identifiers are used on most of our information sheets as Casa Herrera information file locators:

“MINS nnnn” = M(anufacturer’s) IN(formation) S(heets) (4-digit number)

“CHINS nnnn” = C(asa) H(errera) IN(formation) S(heets) (3-digit number)

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240510701
240510702

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050111PS



Product Manual for the

Pre-sheeter

UTZ FOODS

Job # 090101 S.O.# 050111

CHINS 101

Model No.: **PS1039**

Serial No.: **09031168**

Horsepower: **2 HP** Volts: **460 V** Amps: **5 Amps**

INSTALLATION DATE: _____

**If you call for service, the above information
will be requested from you!**

MANUFACTURER'S STATEMENT

It is the responsibility of the customer to comply with local and national codes for installation.

Casa Herrera Inc. offers instruction in safe and proper operation of Casa Herrera machinery. Purchasers of our equipment are strongly urged to consider sending their operating and maintenance personnel to our Pomona plant for "hands-on" training and experience.

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RETAIN THIS MANUAL FOR FUTURE REFERENCE
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SAFETY PRESHEETER

CHINS 137

SAFETY

Presheeter safety is a complex subject. You should always retain records of presheeter conditions, training sessions with cooks and operation and sanitation personnel and should encourage input on safety from the employees at all **regularly scheduled safety meetings**. We could not possibly cover all situations. This would take thousands of pages. We offer the following as a starting point upon which to found your fryer safety program:

ROTATING MACHINERY is dangerous. The human body is unpredictable and can be burned, cut, squashed, torn, pinched and broken in to many ways to count. The key is to keep clear of all points of potential injury and to educate those in the area of the hazard potential. Guards are provided for safety. Removed guards provide opportunity for injury. Worn parts break down...Breakdowns cost money and open opportunities for safety hazards.

PINCH POINTS are the most common hazard on sheeters and pre sheeters. The equipment is shipped with full guarding and these guards must be properly installed at all times. **USE COMMON SENSE.**

SLIP/FALLS are dangerous and food processing plants are wet with oil and water present. Coat all walking surfaces with anti-slip surfaces and post warning signs.

SANITATION CHEMICALS are dangerous. Only personnel properly trained in the use of these chemicals should ever be allowed to use same...including their managers. People using or in the proximity to sanitation chemicals must be dressed in protective clothing, face shields and appropriate breathing protection. They can cause chemical burns to people directly using them or people splashed by them or even touching them...even innocent passers by. They can cause fires, fumes and other chemical reactions. They can damage equipment and facilities. **TRAIN ALL OF YOUR PEOPLE.**

Presheeter machines are Electrical powered. **ELECTRICITY IS DANGEROUS**...especially in a wet environment which is what a food factory is...a wet operation. Always use lock-out procedures and always do daily inspections for possible needed maintenance items. Never work on a motor, switch, or other control component with the power on. Repeat use lock-out procedures.

ENFORCE SAFETY THROUGH AN ENGINEERED PROGRAM DESIGNED FOR “0” ACCIDENTS....DESIGN IT FOR YOUR PLANT.....! ACCIDENTS CAN BE ELIMINATED THROUGH INTELLIGENT WORK

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DO NOT TAKE CHANCES....WORK SAFELY.....Inspect constantly!



Installation...Pre-sheeter

CHINS 151

The Pre-sheeter System is shipped in a single machine assembly with a separate electrical control panel for field wiring to the machine and your power supply. The Masa receiving hopper is removable for sanitation and ease of handling during installation.

1. Set the Pre-sheeter according to the predetermined position over the sheeter, which must be in the proper operation position in the production line. Level the frame. The rolls should be horizontal and parallel to the sheeter rolls for proper operation. Once positioned and double checked, bolt it securely to the frame using appropriate bolts of grade 5 or better grade and either stainless steel or plated steel manufacture to protect against rust and corrosion. Alignment is critical to proper operation of this machine.
2. Once bolted down, grout the picture frame to the floor. Use non-shrink grouting after all bolting is complete. Grouting must be oil, water, and caustic proof.
3. Next mount the Electrical control panel where desired. Mount it so all conduit runs enter from the bottom or side and so it can be operator accessed from adjacent to the sheeter below.
4. The Main Control panel was factory wired and tested with each wire numbered to match the corresponding terminal number in the panel. Fasten and seal the conduits at each end and install the wires per the numbers & schematics provided. Remember to use proper waterproof sealing methods, at all connection points, properly ground the machine and control and properly size the wires according to good practice and electrical codes for a wet location.
5. Install the Hopper on the machine and wire this switch to the Control panel directly.
6. Recheck the entire installation with the sheeter in place and if all is correct install any related equipment such as masa transporting equipment.
7. Once this is done, you can now bump test for rotation and to be certain all is properly assembled and ready to run. Remember, rotation on this machine is rolls turn top towards each other and so do the masa paddles.



THE PRESHEETER SYSTEM:

CHINS 136

This machine is designed to enhance the production of Corn Chips, and Tortilla Chips and Tortillas from corn based material. It is also applied on occasion to other products such as potato based or wheat flour based doughs. Features include doctor blade type dough stripping, textured roll surfaces, infeed anti-bridging paddles, adjustable roll gap and roll speed, and separate drives for the rolls and the paddles.

The Objective: Is to provide a consistent and even flow of product in a preformed sheet to the sheeter where it is finally sized into consistent finished product without the variables caused by uneven sheeter in feeds.

The Anti-bridging paddles: This feature feeds the masa to the rolls with hands off. Masa tends to bridge and extrude or flow away from the pressure area and to slip on top of the rolls. These paddles continuously push the masa down towards the presheeter roll gaps. They farther keep the Masa moving and retard the formation of surface crust. They are removable for cleaning and should operations elect to not use them, the machine can function with them removed and the drive turned off.

The Rolls: The rolls are precision sized and of stainless steel and with textured surfaces to provide traction with the masa. They are geared differentially driven at suitable speed to enhance the feed and stripping of the masa from their surfaces. They are on heavy duty bearing units and the gap is adjustable to control the thickness of the product web feed to the sheeter. One is fitted with a doctor blade to assure product removal and feed.

The Drives: The rolls are driven with their own gearmotor and fitted with adjustable AC Frequency Inverter controllers for speed control which is in fact the product feed rate control method. The paddles also have their own back geared gearmotor drive with separate Inverter controller allowing settings to be altered to suit the many different dough consistencies.

The Hopper: The hopper is of stainless steel and each application is custom designed to suit our customers application needs. Masa can be fed to the hopper in large clumps directly from a mixer, in logs, in a conveyed ribbon, by hand, or most any other method. The top of the hopper will be fitted with a cable emergency stop switch. This switch is absolutely required for safety. (Note: do not allow anything to be inserted into moving machinery such as a tamper)

OPERATION...Pre-Sheeter

CHINS 138

OPERATION

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PADDLE OPERATION:

PADDLES: This machine is fitted with two paddles. The paddle position is not adjustable. They must be installed and their gears properly timed to function properly. Proper timing is when both paddles appear the same when horizontal and viewed from the top. That is, the paddle on the left shaft points the same direction as the paddle on the right shaft does and the paddle on the left shaft points to the gap in between the paddles on the right shaft and visa versa. This is by design as the paddles are designed to knead and push the masa down into the pre sheeting rolls.

PADDLES: are to rotate in the direction in which the dough is pushed down into the "vee" between the sheeter rolls.

ADJUSTABLE PADDLE SPEED: The paddle speed (RPM) is adjustable from the potentiometer in the control panel face. This is the only adjustment to the paddle system other than the option to run with or without same.

PADDLE REMOVAL: Each end of the paddles has slip collars which when loosened and slid towards the center of the machine will allow the removal of the entire paddle shaft.

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PRE SHEETER ROLLS

ROLLS: The rolls are stainless steel and with textured surfaces. One is fixed and one is position adjustable. They are driven from a single direct mounted gearmotor with back-gearing between on each roll for speed a differential drive character. This feature eliminates a dough stripping device (wire or blade) from between the rolls as the dough will automatically transfer to the front roll. From here the dough is stripped by the use of a "doctor blade" dough stripping device with spring type tensioning mechanism.

ROLL GAP ADJUSTMENT: This adjustment should be set at between 1/8" and 3/16" and left alone. The actual gap setting will be determined by the operation director. The objective is to give the final sheeter precisely the correct amount of dough continuously, and to eliminate over or under feed of this sheeter. The ideal set-up would be to see a flowing ribbon down into the sheeter and out of the sheeter without any thin areas in the sheeted product or bunching above the sheeter rolls.

ROLL SPEED ADJUSTMENT: This is where you adjust the volume of masa going to the sheeter. Trying to do so with thickness is very difficult and tricky because an adjustment to one side effects both sides. Once the thickness of the discharge web is even, more or less volume is easily adjusted by more or less roll RPM.



SADDLES: The rolls are fitted with saddles to retard the loss of dough from the sides of the rolls. Saddles must be fitted to the rolls and treated with care to prevent damage or misfit which can result in dough leakage.

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HOPPER:

HOPPER: The hopper is made heavy to withstand the impact of large chunks of dough tumbling about within. It also acts as a safety device to keep personnel out of the machinery. A quick inspection of this machine will show that while it is necessary to have the top open to feed product, it could be considered very hazardous if not treated with common sense and good operation procedures and rules. **This machine is intended to feed dough...not sticks, pipes, people parts or other foreign objects and you are required to keep them out of the hopper.**

Hopper Safety Switch: The top of the hopper is fitted with a cable type rim safety switch. Should a person stumble or fall near this rim, the cable when struck or pulled will cause the machine to instantly stop. If your machine has stopped...do not restart without first inspecting to find the cause of the stoppage and to be certain no person is in harms way.

FEEDING DOUGH: Dough may be fed in large or small chunks, logs, ribbons or any other manner. Try to feed with reasonable care so as not to damage the machine with impact by dropping a large chunk from above or slamming a cart into the side of the equipment. We have units taking dough directly from batch mixers, from conveyors, from extruders, from continuous mixers, by hand feed, etc.

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DOCTOR BLADE:

DOCTOR BLADES: are used to remove dough from a roll in an even sheet. To be even, they must be true and straight and without nicks, gouges, bent spots or other imperfections. They are designed to run against the roll directly and any imperfections will not only give poor performance, but will destroy the roll itself. The doctor blade is renewable and we suggest several be kept in stock at all times. Adjust the spring tension with only enough tension of the blade to the roll to prevent the dough from lifting the blade away from the roll. More tension is destructive to the roll and the blade.

Cleaning the Pre-sheeter

CHINS 140

CLEANING

PRESHEETER CLEANING is IMPORTANT. Presheeter cleaning must be performed as frequently as practical and we recommend that you rinse clean the unit every 4 hours minimum to prevent the build-up of sour masa or sough and the related bacteria. This will remove the majority of the crumbs and residue which can collect in corners and pockets and which will cause rapid degeneration of product quality.

We also recommend that every day end, that the machine be washed with a suitable soap solution as used on your sheeter to farther eliminate the potential of food contamination.

MACHINE CLEANING CAN BE DANGEROUS..! The chemicals used to clean food processing equipment are strong and can and will cause severe burns and injuries to plant personnel. They will also damage machinery components and will damage facilities. Do not take chances.

SANITATION CHEMICALS are dangerous. Only personnel properly trained in the use of these chemicals should ever be allowed to use same...including their managers. Training records must be retained in the personnel office for each person exposed to the chemicals used in plant & equipment cleaning procedures

People using or in the proximity to sanitation chemicals must be dressed in protective clothing, face shields and appropriate breathing protection.

They can cause chemical burns to people directly using them or splashed or even touching them...even innocent passers by.

They can cause fires, fumes and other chemical reactions.

They can damage equipment and facilities.

Because of these facts, it is imperative that you completely understand the nature of the chemicals you are using. We recommend that you demand that your chemical supplier furnish you with suitable containers for safe use and storage, proper Material Safety Data sheets for each chemical, written instructions for the proper use, methods, storage, disposal, and medical treatment information for each chemical. He should also provide you with training materials for both the plant floor and classroom and perhaps he could even be persuaded to conduct the training sessions.

Warning:

In the U.S.A. it is the LAW! You must have all personnel trained and you must have MSDS sheets on all chemicals posted for employee review.

PRESHEETER Cleaning Instructions:



Before starting...wipe down and manually clean all drive motors and electrical boxes exteriors and then wrap them with suitable plastic protective covering.

1. Remove all dough from the machine after it has finished its' run.
2. Remove the sheeter from below the presheeter to prevent wash water from harming this machine. Also splash drape the fore-deck of the oven to prevent water damage there as well.
3. Rinse the machine using a high pressure hose nozzle. Flush residue down to the floor below and then clean the floor.
4. Remove the hopper after disconnecting the safety switch and open the motor safety disconnect switches to prevent accidental starting of the motors.
5. Remove the paddles once per week for detailed cleaning of mating parts and the collars.
6. Open the rolls using the safety release bar...do not turn or adjust the tee handles because you will loose your operational settings.
7. Now use suitable chemicals as recommended by your supplier (Usually something like a strong detergent). This machine is constructed of carbon steel, plated, stainless steel and plastic. Wash the unit totally and rinse thoroughly when done.
8. After the machine has drip dried, and the area surrounding it has been suitably dried, remove the plastic covers, reassemble, test run, and return it to service.

REPEAT:

Remove all of the protective plastic before you quit or restart the machine!



MAINTENANCE Pre-Sheetter

CHINS 139

Maintenance of your Casa Herrera PRESHEETER

Maintenance of a presheetter is like maintenance of any production machine. The better job you do, the lower your operation costs and the more reliable your production output.

To maintain your machine you will need to do the following on a regularly scheduled basis:

1. Inspect daily. Using the look, listen, smell, feel senses to detect potential problems.
2. Demand that the operator keep an observation and notification log for the maintenance group with entries on every deviation from normal performance.
3. Do a complete preventative maintenance inspection every weekend and report findings for scheduling anticipated replacement repairs.
4. Keep the machinery clean...! always. Clean equipment runs cooler and better and lives longer between failures. It is also easier to repair when breakage does occur.
5. Review your parts department and your availability of parts in the proximity of your facility. You cannot repair a machine if you do not have proper tools and parts. If a part is a long lead-time item, it can cost you production down time directly proportionate to that lead-time. Example: Sheetter rolls are usually 2 to 4 weeks lead time...no shortcuts! If your rolls fail, and you have no spares, you are forced to make temporary repairs while the replacement roll is being manufactured and shipped to your plant. Parts are expensive...But...So is Down-Time...!

SPECIFICALLY:

Controls: inspect and keep doors closed & sealed, keep them clean and dry inside and outside. Periodically check for loose terminals, wires, screws, relays and other parts. Repair any leaks at doors or conduit entrances. Keep spare fuses and key parts inventory at the ready. Allow only trained electrical personnel to enter the panel. Cycle safety switches to be certain they perform as intended. Replace immediately if not working perfectly.

Sheetter rolls: Inspect bearings and keep them lubricated properly. Listen for signs of failure using a stethoscope or listening rod. Change them before failure.

Paddles & drives: Inspect for loose retainer bolts monthly. Daily, inspect for loose drive gears and/or damaged bearings. Inspect weekly for worn seals other parts.

Drives: Keep them clean including inside motor cooling fans. Keep breathers clear at all times. Cover when performing sanitation functions to prevent water & chemical damage. Daily check oil levels and inspect for leaking oil seals. Mechanical variable speed units (if so equipped) should be cycled through their speed range daily. Mounting bolts and driven components should be inspected for correctness. Check the wiring connections every 6 months for tightness.

LUBRICATION

	(G) GREASE	(GH) GREASE	(GO 1) GEAR OIL 1			(GO 2) GEAR OIL 2		(CO) CHAIN OIL	(HO) HYDRAULIC OIL
			16 to 50°F	51 to 110°F	111 to 165°F	5 to 77°F	32 to 104°F		
AMERICAN OIL						Spartan EP-150	Spartan EP-220		
ATLANTIC REFINING									
CITIES SERVICE			CITGO Cylinder Oil 460-5	CITGO Cylinder Oil 680-7	CITGO Cylinder Oil 680-7				
CHEVRON						Non-Leaded Gear Compound 150	Non-Leaded Gear Compound 220		
DIXON	Bak-Pruf EG #2 HP Water White	HY Heat GP Graphite Grease							
GULF OIL			Transgear EP-460	Transgear EP-680	Transgear EP-800	EP Lube S-60	EP Lube S-100		
KEYSTONE			WG-A	K-600	WG-Special				
MOBIL OIL			Mobil 600W	Mobil 600W Super	Mobil Extra HECLA	Mobilgear 629	Mobilgear 630		DTE 25 Food Grade
SHELL OIL			Omala 460	Omala 680	Omala 800	Moata Oil 100	Omata Oil 220		
STANDARD OIL									
SUN OIL			Sunep 1090	Sunep 1150	Sunoco Gear Oil 8 AC				
TEXACO			Meropa 220	Vanguard 620	Vanguard 620	Meropa 150	Meropa 220	Waylube 68	Rando HD46 FOOD GRADE
VALVOLINE	Valpex EP2		Light Cylinder Oil	Light Cylinder Oil	Light Cylinder Oil	HP Gear Oil 80W-85-90	HP Gear Oil 80W-85-90	Waylube W-100	

CHINS 142

LUBRICATION: See the chart labeled “LUBRICATION” for generally recommended lubrication. For specific details pertaining to motor-reducers, see the appropriate “MINS” sheets.

WARNING!

LUBRICATING MOVING MACHINERY IS VERY DANGEROUS, AND IS NOT NECESSARY! ALL MACHINERY HAS TO BE SHUT DOWN AND LUBRICATED

CAUTION!

USE ONLY THE LUBRICANTS ASSIGNED BY THE MAINTENANCE SUPERVISOR AND ONLY IN THE PRESCRIBED WAY. DUE TO DIFFERENCES IN CHEMISTRY SOME OILS AND GREASES WILL DESTROY EACH OTHER WHEN USED TOGETHER. THE EFFECT COULD BE DRAMATIC, MUCH LIKE WHAT HAPPENS TO HAND CLEANERS, THE MASS BECOMES LIQUID, WITH NO LUBRICATING QUALITY AT ALL. IF SUCH A THING OCCUR, FLUSH OUT OLD LUBRICANT BEFORE THE NEW LUBRICANT IS USED.

LUBRICATION PARTS:

1. Gears should be greased weekly.
2. Rear roller releases should be lightly greased with food-grade lube weekly.
3. Lube all bearings once a week through fittings provided. Do not force excess lube through seals.
4. Clean the excess grease thoroughly.
5. Lube the sheeter roller gears weekly.